COMPLETE TECHNICAL & FUNCTIONAL DOCUMENTATION

Sastha Sasti / Divya Murti ERP

Enterprise God Statue, Idol & Religious Artifacts ERP & POS System

An all-inclusive architectural and operational guide covering system capabilities, database entity relationships, role permissions, POS checkout engine, inventory tracking, artisan procurement, supplier payments, multi-channel billing, and financial ledgers.

Application Name:	Sastha Sasti / Divya Murti ERP	**Version:**	v2.4.0 (Production Release)
Backend Architecture:	Pure Core PHP 8.2+ MVC Engine	**Database Engine:**	MySQL 8.0+ / MariaDB (InnoDB)
Web Server:	Apache 2.4 (XAMPP Environment)	**Active HTTP Port:**	Port 80 (HTTP) & Port 443 (HTTPS)
Active Database Port:	Port 3306 (MySQL)	**Application Base URL:**	http://localhost/billing/billingsoftwaregod/
Target Industry:	Devotional Statues, Brass, Panchaloha, Marble & Wood	**Documentation Date:**	October 2026

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1. Executive Summary & Solution Architecture

Sastha Sasti / Divya Murti ERP is a specialized enterprise resource planning and point-of-sale solution custom-engineered for god statue manufacturers, sthapathi sculptors, bronze and brass foundries, religious artifact showrooms, and devotional product wholesalers.

Traditional generic retail POS software lacks support for the dimensional attributes, metal compositions (Panchaloha, Brass, Black Granite, Marble, Teak), piece-rate artisan labor markups, and sacred idol taxonomy required by religious

artifact enterprises. This platform closes that gap with a high-throughput, sub-second POS terminal, multi-mode payment collection, automated GST compliance (HSN 9701 / 12% GST), an immutable double-entry stock ledger, and deep supplier artisan accounting.

Architectural Highlights

- **Pure Core MVC**: Lightweight, ultra-fast routing engine without bulky third-party framework overhead.
- **Prepared Statements**: 100% parameter-bound SQL queries preventing SQL injection vulnerabilities.
- **Session & CSRF Security**: Automatic cryptographic tokens validating every mutating POST/AJAX transaction.
- **Client-Side Resilience**: Held carts persisted in local browser storage across hardware restarts.

Business Capabilities

- **Instant A4 / Thermal Billing**: Print GST tax invoices or 80mm/58mm thermal receipts instantly.
- **Artisan Foundries P&L**: Track piece-rate purchase costs, markups, and supplier debt in real time.
- **Granular Role Security**: Independent interfaces tailored for Super Admins down to dedicated POS Cashiers.
- **Forensic Audit Trail**: Every critical login, update, sale, and stock adjustment is logged with IP and timestamp.

2. Infrastructure, Server Stack & Port Configurations

The software is configured to operate in high-availability environments and local intranet deployments via XAMPP Apache and MySQL services.

SERVICE COMPONENT	PORT	PROTOCOL	SOFTWARE & ENGINE	CONFIGURATION / PURPOSE
Apache Web Server (HTTP)	`80`	TCP / HTTP	Apache 2.4.58 (Win64)	Main application HTTP endpoint, serving static assets, MVC routing, and API calls.
Apache Secure Server (HTTPS)	`443`	TCP / HTTPS	Apache OpenSSL 3.1.3	Encrypted transport layer for production SSL certificates.
MySQL Database Engine	`3306`	TCP / MySQL	MySQL 8.0 / MariaDB 10.4	Relational database storage (`god_statue_erp`) with strict foreign keys & transactions.
PHP Runtime Environment	In-Process	FastCGI / Module	PHP 8.2.12	Server-side scripting with extensions: `mysqli`, `pdo_mysql`, `mbstring`, `openssl`, `json`.

> **Application Access Endpoints**
>
> **Base Application URL:** `http://localhost/billing/billingsoftwaregod/`
> **Direct Login Endpoint:** `http://localhost/billing/billingsoftwaregod/login`
> **Database Management (phpMyAdmin):** `http://localhost/phpmyadmin/`

3. Role-Based Access Control (RBAC) & User Accounts

The application enforces strict Role-Based Access Control (RBAC). Access to every route and controller action is guarded by middleware checking against the `permissions` and `role_permissions` database tables. Additionally, custom overrides can be assigned to individual users via `user_permissions`.

ROLE NAME	USERNAME	PASSWORD	PRIMARY RESPONSIBILITIES & ACCESS SCOPE
SUPER ADMIN	`superadmin`	`admin123`	Unrestricted full access across all modules, configuration settings, user creation, role editing, and audit trail logs.
STORE ADMIN	`rajesh`	`admin123`	Operations management, product catalog maintenance, customer approvals, purchase order authorizations, and financial analytics.
BILLING HEAD	`anand`	`admin123`	Supervision of counter cashiers, invoice cancellations, customer credit approvals, and daily POS register reconciliation.
POS CASHIER	`priya`	`admin123`	Streamlined POS terminal, fast barcode scanning, bill creation, payment processing, and A4/thermal invoice printing.
CHIEF ACCOUNTANT	`venkatesh`	`admin123`	Cash & bank ledgers, supplier payments, expense voucher approval, tax summaries, customer outstanding dues, and P&L statements.
WAREHOUSE LEAD	`murugan`	`admin123`	Stock inward verification, physical stock counts, inventory adjustments, damaged goods write-offs, and procurement receipt.

4. Complete Database Schema (28 Tables & Relationships)

The system runs on the `god_statue_erp` database featuring 28 relational tables enforced by InnoDB foreign keys, primary keys, and indexes for high-concurrency read/write operations.

TABLE NAME	PRIMARY KEY	FOREIGN KEYS & RELATIONSHIPS	DESCRIPTION & BUSINESS PURPOSE
`departments`	`id`	-	Organizational departments (Billing, Sales, Warehouse, Accounts).
`roles`	`id`	-	System security roles with unique slugs and permission sets.
`permissions`	`id`	-	Granular capabilities mapped to modules (e.g., `billing.pos`, `products.edit`).
`role_permissions`	`(role_id, permission_id)`	`roles.id`, `permissions.id`	Many-to-many relationship mapping roles to specific permissions.
`users`	`id`	`roles.id`, `departments.id`	Staff members, encrypted bcrypt passwords, avatar paths, last login IPs.
`user_permissions`	`(user_id, permission_id)`	`users.id`, `permissions.id`	Custom user-specific permission grants or explicit revocations.
`categories`	`id`	-	High-level classifications (Brass Statues, Marble Idols, Wooden Carvings).
`subcategories`	`id`	`categories.id`	Deity classifications (Ganesha, Shiva, Parvathi, Murugan, Perumal, Lakshmi).
`products`	`id`	`categories.id`, `subcategories.id`, `suppliers.id`	Full idol catalog, dimensions (H/W/D/Wt), piece rate, markup, selling price, barcode.
`product_images`	`id`	`products.id`	Multiple gallery images for each statue with primary image flags.
`customers`	`id`	-	Customer CRM, retail, wholesale, temple trusts, credit limits, outstanding balances.
`suppliers`	`id`	-	Sthapathis, artisans, foundries, bank IFSC, account numbers, GST, balances.
`supplier_payments`	`id`	`suppliers.id`	Summary payment state (Paid, Unpaid, Partially Paid) and total paid amount per vendor.
`supplier_payment_history`	`id`	`suppliers.id`, `users.id`	Immutable audit history of all payment disbursements made to artisans/suppliers.
`purchases`	`id`	`suppliers.id`, `users.id`	Procurement purchase orders, total amounts, taxes, approval status.

TABLE NAME	PRIMARY KEY	FOREIGN KEYS & RELATIONSHIPS	DESCRIPTION & BUSINESS PURPOSE
`purchase_items`	`id`	`purchases.id`, `products.id`	Line items for purchases specifying received quantities and cost prices.
`invoices`	`id`	`customers.id`, `users.id`	Sales invoices, tax breakdowns (CGST/SGST/IGST), discount, grand total, status.
`invoice_items`	`id`	`invoices.id`, `products.id`	Line items on sales invoices with snapshot of idol attributes, unit price, tax.
`sales_returns`	`id`	`invoices.id`, `customers.id`, `users.id`	Sales return records, refund amounts, refund methods, reason notes.
`sales_return_items`	`id`	`sales_returns.id`, `products.id`	Individual products returned and restocked back into the warehouse.
`inventory_transactions`	`id`	`products.id`, `users.id`	Immutable transaction ledger: stock in, stock out, sale, restock, damage, adjustment.
`accounts`	`id`	-	Chart of accounts (Main Cash Register, HDFC Current, SBI Merchant, Petty Cash).
`payments`	`id`	`accounts.id`, `customers.id`, `suppliers.id`	Double-entry payment receipts and voucher records with bank UTR/cheque refs.
`expense_categories`	`id`	-	Expense types (Foundry Freight, Temple Rent, Electricity, Packaging, Wages).
`expenses`	`id`	`expense_categories.id`, `accounts.id`	Expense vouchers with payee info, payment method, bill attachments, approvals.
`account_transactions`	`id`	`accounts.id`, `users.id`	Financial journal debit/credit entries updating running account balances.
`settings`	`id`	-	System key-value configurations (Company name, address, GSTIN, default tax rate).
`activity_logs`	`id`	`users.id`	Forensic audit log recording all user logins, updates, deletes, and printings.

5. Core Application Modules & Feature Inventory

5.1 Executive Dashboard

The executive dashboard aggregates real-time business health metrics across the enterprise:

- **Financial KPI Cards**: Displays Total Sales Today, Month-to-date Revenue, Total Invoices Generated, Paid Amount Collected, and Current Outstanding Customer Dues.
- **Inventory Critical Alerts**: Live monitoring of idols reaching reorder thresholds with immediate "Reorder" action triggers.
- **Dynamic Trend Charts**: Integrated Chart.js visualizations demonstrating daily and weekly sales trajectory.
- **Deity Popularity Index**: Ranks highest-velocity statues (e.g., 12-inch Brass Ganesha, Panchaloha Nataraja) by sales volume.

5.2 Sacred POS Billing Terminal (/billing, /pos)

Engineered for rapid counter operations during high-traffic festival seasons and temple peak hours:

- **Lightning Search & Quick Categories**: Instant sub-300ms fuzzy search by deity name, SKU, material, or barcode with quick category filter tabs.
- **Barcode Scanner Integration**: Automatically intercepts USB/Bluetooth barcode scanner inputs; typing or scanning barcodes automatically pulls up items with stock validation.
- **Cart Management & Dynamic Modifiers**: Real-time quantity steppers, item-level discounts, stock quantity threshold enforcement, and removal triggers.
- **Bill-Level GST Customization**: Billing operators can select custom GST rates (e.g., 0%, 5%, 12%, 18%) or apply item defaults, dynamically updating taxable breakdowns.
- **Multi-Channel Payment Engine**: Supports Cash, UPI (with QR code simulation), Credit/Debit Cards, Bank Direct Transfer, and Customer Ledger Credit. Also supports split-mode settlements.
- **Held Carts (Suspend & Resume)**: Allows cashiers to suspend multiple active carts to attend to other customers, saving them safely in the browser's localStorage.
- **Instant Dual-Format Print Triggers**: Instant generation and printing of:
  - **A4 GST Tax Invoices**: Formal tax document complete with company GSTIN, HSN codes, CGST/SGST decomposition, customer address, and authorized signature section.
  - **Thermal Receipts (80mm & 58mm)**: Compact receipt format designed for rapid thermal receipt printers with company header, item list, and barcode.

5.3 Customer Invoices Register (/customer-invoices)

A unified register for reviewing, filtering, and managing completed and pending sales bills:

- **Advanced Multi-Criteria Filtering**: Filter by invoice number, customer phone number, payment status (Paid, Partial, Unpaid), invoice status, and custom date spans.
- **Key Financial Metrics**: Immediate top-level display of Total Invoiced Revenue, Net Collected Cash, and Total Outstanding Credit Receivables.
- **Actions & Operations**: View interactive invoice details modal, reprint A4 invoices, reprint thermal slips, export to PDF, cancel invoices with audit notes, and accept due payments.

5.4 Product Management & Statue Catalog (/products)

The product catalog features an interactive four-pillar tabbed interface designed for seamless idol management:

1. Supplier Details Tab

Manages artisan sculptors and foundries. Provides a real-time table of all registered suppliers, contact information, product count, payment status, and an inline modal to register suppliers along with initial product associations.

2. Product Details Tab

Full catalog entries for statues. Captures specific attributes: Deity Name, Material, Dimensions (Height, Width, Depth, Weight), Finish Type (Antique, Gold Plated, Polished), Piece Rate, Markup %, Selling Price, Wholesale Price, and Low-stock limits.

3. Barcode Details Tab

Automated barcode label generation. Configures SKU tracking parameters, generates scannable Code128 barcodes, and provides a multi-label print preview ready for thermal sticker printers.

4. Invoice Details Tab

Tracks supplier purchase transactions, invoice numbers, total purchase value, paid amount, and payment status (Paid, Unpaid, Partially Paid), with instant payment update modals.

- **Bulk Data Exchange**: Features one-click CSV/Excel Catalog Export, Bulk CSV Import with validation, and a downloadable standard CSV template for fast inventory initialization.

5.5 Categories & Subcategories (/categories, /subcategories)

- **Hierarchical Deities Taxonomy**: Organizes inventory into primary categories (e.g., Brass Statues, Marble Murtis, Wooden Artifacts) and deity subcategories (Lord Ganesha, Lord Shiva, Goddess Lakshmi, Lord Murugan, Perumal, Temple Bells).
- **Category Management**: Unique category codes, descriptions, cover image upload, and bulk CSV Import/Export.

5.6 Stock Control & Inventory Transaction Ledger (/inventory)

- **Immutable Double-Entry Ledger**: Every single piece mutation is strictly recorded in `inventory_transactions` with reference type, previous stock, quantity change, new stock, and operator user ID.
- **Transaction Classification**: Distinguishes between `stock_in`, `stock_out`, `sale`, `purchase`, `sales_return`, `adjustment`, `damage`, and `transfer`.
- **Stock Adjustment Modal**: Allows authorized staff to perform physical audit counts and register adjustments with mandatory reason documentation.

5.7 Artisans, Foundries & Supplier Payments (/suppliers, /billing/invoices)

- **Artisan Profiles**: Maintains master records of master sculptors (Sthapathis), bronze foundries, and woodcarvers, including banking IFSC codes, account numbers, and GST credentials.
- **Supplier Payment Tracking**: Integrated tracking through `supplier_payments` and `supplier_payment_history` tables.
- **Payment Modal & History**: Enables accountants to record partial or full payments, choose payment mode (Cash, Bank Transfer, UPI, Cheque), record transaction reference IDs, and view chronological disbursement histories.

5.8 Purchase Procurement Orders (/purchases)

- **Procurement Lifecycle**: Supports multi-tier purchase workflow: `draft` → `pending` → `approved` → `received`.
- **Automatic Stock Inward**: Approving or marking a purchase as received automatically increments inventory in the catalog and creates corresponding ledger entries in `inventory_transactions`.

5.9 Customer CRM & Ledgers (/customers)

- **Customer Categorization**: Profiles categorized into Retail Walk-in, Wholesale Distributors, and Temple Trusts / Devasthanams.
- **Credit & Ledger Control**: Enforces maximum credit limits and tracks running outstanding ledger balances across billing cycles.

5.10 Accounts, Cash Book & Bank Registers (/accounts)

- **Chart of Accounts**: Dual registers managing Physical Counter Cash and Bank Current Accounts (e.g., HDFC, SBI).
- **Receipts & Disbursements**: Centralized recording of customer payment collections against unpaid invoices and supplier payouts.
- **Running Balance Tracking**: Real-time ledger entries in `account_transactions` with credit/debit validation.

5.11 Expense Vouchers (/expenses)

- **Voucher Generation**: Creation of expense vouchers across categories: Showroom Rent, Freight & Logistics, Electricity, Packaging, Wages, and Workshop Advances.
- **Payment Reconciliation**: Automatic debiting from the specified Cash or Bank account upon voucher approval.

5.12 Sales Returns & Restocking (/sales-returns)

- **Invoice Lookup Engine**: Cashiers can search the original invoice number to automatically populate purchased items and prices.
- **Restocking & Refunds**: Returning items immediately restocks them back into warehouse inventory and generates a cash/UPI/credit note refund voucher.

5.13 Analytics, Valuation & P&L Reports (/reports)

- **Sales Analytics Report**: Visual breakdown of daily, weekly, and monthly sales trends, gross revenue, discount totals, and tax collections.
- **Stock Valuation Report**: Analyzes real-time warehouse inventory value at cost price versus retail selling price, highlighting dead stock and fast-moving idols.
- **Financial P&L Statement**: Real-time Profit & Loss statement summarizing Sales Income, Cost of Goods Sold (COGS), Operating Expenses, and Net Operating Margins.

5.14 System Settings & Forensic Activity Audit (/settings, /audit)

- **Company Branding**: Configures business legal name, branch address, contact numbers, GSTIN, currency symbol (₹), and invoice number prefixes.
- **Forensic Activity Trail**: Immutable log table recording user ID, action type, module, record ID, IP address, user agent, and timestamp for complete internal compliance.

6. Complete HTTP Routing & API Reference

The table below provides a complete reference of all system endpoints defined in `routes/web.php`:

HTTP METHOD	URI PATTERN	CONTROLLER & ACTION	PERMISSION REQUIRED
GET	/, /dashboard	DashboardController@index	Authenticated
GET, POST	/login, /logout	AuthController@login / logout	Public
GET, POST	/forgot-password	AuthController@forgotPassword	Public
GET, POST	/profile	UserController@profile	Authenticated
GET, POST	/users, /users/create	UserController@index / store	users.manage
POST	/users/edit/{id}, /users/delete/{id}	UserController@update / delete	users.manage
GET, POST	/roles, /roles/create	RoleController@index / store	roles.manage
GET, POST	/categories, /categories/create	CategoryController@index / store	categories.view / manage
POST, GET	/categories/import, /categories/template	CategoryController@import / template	categories.manage
GET, POST	/subcategories, /subcategories/create	SubCategoryController@index / store	categories.view / manage
GET	/products, /products/api-data	ProductController@index / apiData	products.view
GET, POST	/products/create, /products/edit/{id}	ProductController@create / update	products.manage
POST	/products/save-detail	ProductController@saveDetail	products.manage
GET, POST	/products/export, /products/import	ProductController@export / import	products.manage
GET	/billing, /pos	BillingController@pos	billing.pos
GET	/billing/search-products	BillingController@searchProducts	billing.pos
POST	/billing/scan	BillingController@scanBarcode	billing.pos
POST	/billing/checkout	BillingController@checkout	billing.create
GET	/customer-invoices	BillingController@customerInvoices	billing.view
GET	/billing/invoices	BillingController@invoices	billing.view
GET	/billing/invoice/{id}/print	BillingController@printInvoice	billing.view
POST	/billing/invoices/update-payment	BillingController@updateSupplierPayment	billing.view

HTTP METHOD	URI PATTERN	CONTROLLER & ACTION	PERMISSION REQUIRED
GET	/billing/invoices/supplier/{id}	BillingController@supplierDetail	billing.view
GET, POST	/purchases, /purchases/create	PurchaseController@index / store	purchases.view / manage
POST	/purchases/approve/{id}	PurchaseController@approve	purchases.manage
GET, POST	/suppliers, /suppliers/create	SupplierController@index / store	suppliers.view / manage
GET, POST	/customers, /customers/create	CustomerController@index / store	customers.view / manage
GET, POST	/sales-returns, /sales-returns/create	SalesReturnController@index / store	billing.returns
GET	/accounts, /accounts/payments	AccountController@index / payments	accounts.view
POST	/accounts/customer-payment	AccountController@recordCustomerPayment	accounts.manage
POST	/accounts/supplier-payment	AccountController@recordSupplierPayment	accounts.manage
GET, POST	/expenses, /expenses/create	ExpenseController@index / store	expenses.view / manage
GET	/reports/sales, /reports/inventory	ReportController@sales / inventory	reports.sales / inventory
GET	/reports/financial	ReportController@financial	reports.financial
GET, POST	/settings, /settings/update	SettingsController@index / update	settings.view / manage
GET	/audit	AuditController@index	audit.view

7. Operational Workflows & Standard Operating Procedures (SOP)

7.1 Daily Counter POS Billing Workflow

1. **Cashier Sign-in**: Log in using assigned POS cashier credentials (e.g., `priya` / `admin123`). The interface opens directly to the Sacred POS Billing Terminal.
2. **Item Addition**: Point the barcode scanner at idol labels or search by deity name / size. Items instantly populate the active cart with default selling price and stock verification.
3. **Customer Identification**: For retail walk-ins, leave as Counter Customer. For regular temple patrons or wholesalers, enter their mobile number to auto-populate or create their profile.
4. **Tax & Discount Selection**: Adjust item discount or overall bill discount if authorized. Select custom GST tax % or retain standard 12% GST.
5. **Payment & Checkout**: Click checkout, select tender mode (Cash, UPI, Card, Bank Transfer, Ledger Credit), enter amount paid, and confirm transaction.
6. **Print Invoicing**: The print dialog triggers automatically. Choose **A4 GST Tax Invoice** for institutional orders or **Thermal Receipt** for retail counter patrons.

7.2 Artisan Sourcing & Supplier Settlement Workflow

1. **New Artisan Registration**: Open *Artisans & Foundries* or the *Supplier Details Tab* in Product Management. Record sculptor name, phone number, foundry location, bank IFSC, and account number.
2. **Product Association & Costing**: Record received idols with piece-rate labor cost, markup percentage, and calculated selling price.
3. **Payment Disbursement**: In *Supplier Purchase & Payment Management* (`/billing/invoices`), click **Update Payment** for the supplier. Record the disbursed amount, payment method (Bank/UPI/Cash), date, and reference UTR.
4. **Audit Verification**: The system immediately updates the supplier's balance, status (Paid / Partial / Unpaid), and records an immutable log in `supplier_payment_history` .

7.3 Day-End Cash Balancing & Financial Reconciliation

1. **Sales Verification**: Access *Customer Invoice Details* (`/customer-invoices`) and filter by today's date to verify total invoiced volume and collected cash.
2. **Cash Register Balancing**: Open *Accounts & Cash Book* (`/accounts`). Match physical drawer cash against the calculated running balance in the Cash register.
3. **Expense Reconciliation**: Verify that all petty cash payments made during the day have corresponding approved vouchers in *Expense Vouchers* (`/expenses`).
4. **Activity Audit Review**: Store Admin or Super Admin reviews the *Audit Trail* (`/audit`) to verify any invoice cancellations or stock adjustments performed during the business day.

> **Database Backup Routine**
>
> It is strongly recommended to run daily database backups. An initial reference backup is maintained at `god_statue_erp_backup.sql` . To create a fresh daily snapshot via command line:
>
> ```
> mysqldump -u root god_statue_erp > "C:\xampp\htdocs\billing\billingsoftwaregod\backup_$(Get-Date -Format 'yyyyMMdd').sql"
> ```